The Uber Assumption and its Random Self-Reducibility in Non-Bilinear and Type 1, 2, 3 Bilinear Groups

Abstract

We fix notation for prime-order groups without a pairing and for bilinear groups in each of the three Galbraith–Paterson–Smart types, formally define the Uber assumption in all four settings, and treat random self-reducibility in each. A single generic-computability lemma isolates, per setting, the space of exponents a reduction can realize; a single affine-reparametrization theorem then yields random self-reducibility whenever the source data are stable under the reparametrization and the challenge polynomial transforms covariantly. The four settings differ only through the product structure the pairing (and, in Type 2, the homomorphism ψ) makes available, and this difference is exactly what enlarges or shrinks the self-reducible subclass. We give a worked self-reducible assumption in each setting and exhibit a target that is self-reducible in Type 2 but not, by any affine reparametrization, in Type 3.

1 Introduction

The discrete-logarithm problem is random self-reducible by the shift $g^x \mapsto g^x g^r$ [1]; the answer for a worst-case instance is recovered from an average-case solver by subtracting r. This note carries the underlying idea, inject a uniform reparametrization of the secret and undo it on the answer, across the standard group settings and states it uniformly. Throughout, p is a prime and all groups have order p. We use implicit (bracket) notation in the style of [5].

2 Group settings

2.1 Non-bilinear groups

Definition 2.1 (prime-order group)**.** A *(non-bilinear) group generator* outputs $\mathcal{G} = (p, \mathbb{G}, g)$ where $\mathbb{G}$ is a cyclic group of prime order p, g a generator, and the group law and equality are computable in probabilistic polynomial time. For $a \in \mathbb{Z}_p$ write $[a] := g^a$, so $[a] \cdot [b] = [a + b]$ and $[a]^c = [ca]$. There is no pairing.

2.2 Bilinear groups and the GPS taxonomy

Definition 2.2 (bilinear group)**.** A *bilinear group generator* outputs $\mathcal{BG} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, g_1, g_2)$ where $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ are cyclic of prime order p, g_1, g_2 generate $\mathbb{G}_1, \mathbb{G}_2$, all group laws and equalities are efficient, and $e \colon \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ is an efficiently computable, non-degenerate bilinear map. For $a \in \mathbb{Z}_p$ write $[a]_1 := g_1^a$, $[a]_2 := g_2^a$, and $[a]_T := e(g_1, g_2)^a$, so that

$$e\big([a]_1, [b]_2\big) = [ab]_T, \qquad [a]_i \cdot [b]_i = [a+b]_i, \qquad [a]_i^c = [ca]_i.$$

Following Galbraith, Paterson, and Smart [2], bilinear groups come in three types, distinguished by the efficiently computable isomorphisms between $\mathbb{G}_1$ and $\mathbb{G}_2$:

Type 1. $\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}$ and e is symmetric, $e(u,v) = e(v,u)$. One source group.

Type 2. $\mathbb{G}_1 \neq \mathbb{G}_2$, with an efficiently computable isomorphism $\psi\colon \mathbb{G}_2 \to \mathbb{G}_1$ (so $\psi([a]_2) = [a]_1$), but none known in the reverse direction.

Type 3. $\mathbb{G}_1 \neq \mathbb{G}_2$, with no efficiently computable isomorphism in either direction.

Type 3 is the most efficient and most used in practice; Type 1 is the most permissive algebraically. The only structural difference for what follows is which group elements can be moved between $\mathbb{G}_1$ and $\mathbb{G}_2$, and hence which products the pairing can form.

3 The generic-computability calculus

Fix a setting and an instance consisting of encodings of finite polynomial tuples $R \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_1$, $S \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_2$, and $T \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_T$ (in a single-group setting only R and, if applicable, T occur). Each tuple is assumed to contain the constant 1, since the generators are public. For $\vec{X} = (X_1, \dots, X_n)$ and a secret $\vec{x} \in \mathbb{Z}_p^n$, the instance is the collection of $[r_i(\vec{x})]_1$, $[s_j(\vec{x})]_2$, $[t_k(\vec{x})]_T$.

For a group $\gamma \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$ (or $\mathbb{G}$) let $\mathcal{L}_\gamma \subseteq \mathbb{Z}_p[\vec{X}]$ denote the polynomials q for which $[q(\vec{x})]_\gamma$ is computable in probabilistic polynomial time from the instance and public parameters, uniformly in $\vec{x}$. Write $A \cdot B = \{ab : a \in A,\ b \in B\}$ and let $\langle\cdot\rangle$ be $\mathbb{Z}_p$-linear span.

Lemma 3.1 (computable exponents). *With the conventions above,*

$$\begin{aligned}
\textit{non-bilinear:}\quad & \mathcal{L}_{\mathbb{G}} = \langle R\rangle;\\
\textit{Type 1 } (\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G},\ \textit{source tuple } R):\quad & \mathcal{L}_{\mathbb{G}} = \langle R\rangle, \quad \mathcal{L}_{\mathbb{G}_T} = \langle R \cdot R \,\cup\, T\rangle;\\
\textit{Type 2 } (\psi\colon \mathbb{G}_2 \to \mathbb{G}_1):\quad & \mathcal{L}_1 = \langle R \cup S\rangle, \quad \mathcal{L}_2 = \langle S\rangle, \quad \mathcal{L}_T = \langle (R \cup S) \cdot S \,\cup\, T\rangle;\\
\textit{Type 3:}\quad & \mathcal{L}_1 = \langle R\rangle, \quad \mathcal{L}_2 = \langle S\rangle, \quad \mathcal{L}_T = \langle R \cdot S \,\cup\, T\rangle.
\end{aligned}$$

The inclusions "$\supseteq$" hold unconditionally: every listed polynomial's encoding is computable by group operations and pairings. The reverse inclusions hold in the generic bilinear group model, where these are exactly the computable encodings [3, 4].

Justification. The group law realizes all $\mathbb{Z}_p$-linear combinations, giving $\langle\cdot\rangle$ in each group. The pairing sends a computable $\mathbb{G}_1$-exponent $a \in \mathcal{L}_1$ and a computable $\mathbb{G}_2$-exponent $b \in \mathcal{L}_2$ to $ab \in \mathcal{L}_T$; pairing against the generator ($b = 1$ or $a = 1$) shows $\mathcal{L}_1, \mathcal{L}_2 \subseteq \mathcal{L}_T$. In Type 1 both pairing arguments range over $\langle R\rangle$, producing all products $R \cdot R$. In Type 2, ψ moves each $\mathbb{G}_2$ encoding into $\mathbb{G}_1$, so $\mathcal{L}_1$ gains S, and the pairable products become $\mathcal{L}_1 \cdot \mathcal{L}_2 = (R \cup S) \cdot S$; the missing products $R \cdot R$ would require an element of $\mathbb{G}_2$ with an R-exponent, which ψ cannot supply. In Type 3 neither group maps to the other, so $\mathcal{L}_1 = \langle R\rangle$,

$\mathcal{L}_2 = \langle S \rangle$, and only cross products $R \cdot S$ arise. The generic-model converse is the master lemma of [3, 4]. $\square$

The single distinction across the four rows is the set of $\mathbb{G}_T$-products: none (no pairing), $R \cdot R$ (Type 1), $R \cdot S \cup S \cdot S$ (Type 2), $R \cdot S$ (Type 3). Everything below is a consequence of this.

4 The Uber assumption

Definition 4.1 (bilinear Uber problem). Fix a bilinear setting, source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$), each containing 1, a challenge polynomial $f \in \mathbb{Z}_p[\vec{X}]$, and a challenge group $ch \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$; write $\mathcal{L}_\star \coloneqq \mathcal{L}_{ch}$ (Lemma 3.1). For a secret $\vec{x} \in \mathbb{Z}_p^n$ the instance is

$$\mathsf{Inst}(\vec{x}) = \big(\{[r_i(\vec{x})]_1\}_i,\ \{[s_j(\vec{x})]_2\}_j,\ \{[t_k(\vec{x})]_T\}_k\big).$$

- *Computational* $(R, S, T, f; ch)$-Uber: given $\mathsf{Inst}(\vec{x})$, output $[f(\vec{x})]_\star$.
- *Decisional* $(R, S, T, f; ch)$-Uber: given $\mathsf{Inst}(\vec{x})$ and Z, decide whether $Z = [f(\vec{x})]_\star$ (real) or $Z \leftarrow_\$ ch$ (random).

The *average case* draws $\vec{x} \leftarrow_\$ \mathbb{Z}_p^n$; the *worst case* quantifies over all $\vec{x}$. For a decision algorithm $\mathcal{A}$ put

$$\delta(\mathcal{A}; \vec{x}) = \Pr\big[\mathcal{A}(\mathsf{Inst}(\vec{x}), [f(\vec{x})]_\star) = 1\big] - \Pr\big[\mathcal{A}(\mathsf{Inst}(\vec{x}), U) = 1\big], \quad U \leftarrow_\$ ch,$$

and $\delta_{avg}(\mathcal{A}) = \mathbb{E}_{\vec{x} \leftarrow_\$ \mathbb{Z}_p^n}[\delta(\mathcal{A}; \vec{x})]$; for a search algorithm put $\mathsf{Succ}(\mathcal{A}; \vec{x}) = \Pr[\mathcal{A}(\mathsf{Inst}(\vec{x})) = [f(\vec{x})]_\star]$ and $\mathsf{Succ}_{avg}(\mathcal{A}) = \mathbb{E}_{\vec{x}}[\mathsf{Succ}(\mathcal{A}; \vec{x})]$. The assumption asserts that δ_{avg} (resp. Succ_{avg}) is negligible for all probabilistic polynomial-time $\mathcal{A}$.

The four settings are Definition 4.1 with the computable spaces of Lemma 3.1. We record each explicitly.

Definition 4.2 (non-bilinear Uber). Setting $\mathcal{G} = (p, \mathbb{G}, g)$; source tuple R (in $\mathbb{G}$, containing 1); challenge $f \in \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}$. Instance $\mathsf{Inst}(\vec{x}) = \{[r_i(\vec{x})]\}_i$; goal, compute or distinguish $[f(\vec{x})]$. Computable space $\mathcal{L}_\star = \mathcal{L}_\mathbb{G} = \langle R \rangle$; no pairing, hence no products.

Definition 4.3 (Type-1 Uber). Setting $\mathcal{BG}$ of Type 1, $\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}$; source tuple R (in $\mathbb{G}$), target tuple T (in $\mathbb{G}_T$), each containing 1; challenge f in $ch \in \{\mathbb{G}, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_\mathbb{G} = \langle R \rangle$ and $\mathcal{L}_{\mathbb{G}_T} = \langle R \cdot R \cup T \rangle$.

Definition 4.4 (Type-2 Uber). Setting $\mathcal{BG}$ of Type 2 with $\psi\colon \mathbb{G}_2 \to \mathbb{G}_1$; source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$); challenge f in $ch \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_1 = \langle R \cup S \rangle$, $\mathcal{L}_2 = \langle S \rangle$, $\mathcal{L}_T = \langle (R \cup S) \cdot S \cup T \rangle$.

Definition 4.5 (Type-3 Uber). Setting $\mathcal{BG}$ of Type 3; source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$); challenge f in $ch \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_1 = \langle R \rangle$, $\mathcal{L}_2 = \langle S \rangle$,

$\mathcal{L}_T = \langle R \cdot S \cup T \rangle$.

Remark 4.6 (hardness). The master theorem of [3, 4] states, informally, that the decisional $(R, S, T, f; \mathrm{ch})$-Uber problem is hard in the generic bilinear group model if and only if $f \notin \mathcal{L}_\star$, that is, the challenge exponent is linearly independent of the computable ones. Random self-reducibility is a separate property and does not require hardness; the two interact through Lemma 3.1, as discussed in Section 10.

5 An affine random-self-reducibility theorem

Let $\mathrm{AGL}_n(\mathbb{Z}_p)$ be the affine group of maps $\tau_{A,\vec{b}}\colon \vec{x} \mapsto A\vec{x} + \vec{b}$ with $A \in \mathrm{GL}_n(\mathbb{Z}_p)$, acting on polynomials by $(q \circ \tau)(\vec{X}) = q(A\vec{X} + \vec{b})$. The following theorem is setting-agnostic: it refers only to the computable spaces of Lemma 3.1.

Theorem 5.1 (master reduction). *Consider any of the four settings, an instance with source tuples R, S, T and challenge $(f; \mathrm{ch})$, and computable spaces $\{\mathcal{L}_\gamma\}$ with $\mathcal{L}_\star = \mathcal{L}_{\mathrm{ch}}$. Suppose there is an efficiently sampleable distribution $\mathcal{D}$ on a subgroup $\mathcal{T} \leqslant \mathrm{AGL}_n(\mathbb{Z}_p)$ such that, for every $\tau \in \mathcal{T}$,*

(C1) $r_i \circ \tau \in \mathcal{L}_1$, $s_j \circ \tau \in \mathcal{L}_2$, *and* $t_k \circ \tau \in \mathcal{L}_T$ *for all* i, j, k *(with* $\mathcal{L}_1 = \mathcal{L}_G$ *in the single-group settings);*

(C2) $f \circ \tau = \lambda_\tau f + h_\tau$ *for a scalar* $\lambda_\tau \in \mathbb{Z}_p^\times$ *and a polynomial* $h_\tau \in \mathcal{L}_\star$*, both computable from* τ *and the parameters;*

and, moreover,

(C3) $\tau(\vec{x})$ *is uniform on* $\mathbb{Z}_p^n$ *for every fixed* $\vec{x}$ *when* $\tau \leftarrow \mathcal{D}$.

Then the computational and decisional $(R, S, T, f; \mathrm{ch})$-Uber problems are random self-reducible: there is a probabilistic polynomial-time reduction $\mathcal{B}^{\mathcal{A}}$, using one draw from $\mathcal{D}$ and one oracle call, with $\mathsf{Succ}(\mathcal{B}; \vec{x}_0) = \mathsf{Succ}_{\mathrm{avg}}(\mathcal{A})$ and $\delta(\mathcal{B}; \vec{x}_0) = \delta_{\mathrm{avg}}(\mathcal{A})$ for every $\vec{x}_0 \in \mathbb{Z}_p^n$.

Proof. On input $\mathsf{Inst}(\vec{x}_0)$ (and a challenge Z in the decisional case) the reduction proceeds as follows.

Reparametrize. Draw $\tau \leftarrow \mathcal{D}$ and set $\vec{x}' := \tau(\vec{x}_0)$, which is unknown to $\mathcal{B}$ but uniform on $\mathbb{Z}_p^n$ by (C3).

Rebuild the instance. By (C1), each rebuilt exponent lies in the computable space of its group, so by the "$\supseteq$" part of Lemma 3.1 its encoding is computable from $\mathsf{Inst}(\vec{x}_0)$. Concretely, writing $r_i \circ \tau = \sum_a c_a g_a$ as a $\mathbb{Z}_p$-combination of the generators of $\mathcal{L}_1$, one has $[r_i(\vec{x}')]_1 = [(r_i \circ \tau)(\vec{x}_0)]_1 = \prod_a [g_a(\vec{x}_0)]_1^{c_a}$, and similarly for S (using ψ in Type 2 to realize any S-exponent in $\mathbb{G}_1$) and for T (using pairings $e([\cdot]_1, [\cdot]_2)$ to realize products). This yields $\mathsf{Inst}(\vec{x}')$, distributed exactly as a fresh average-case instance.

Transport the challenge. By (C2) compute λ_τ, h_τ and the element $[h_\tau(\vec{x}_0)]_\star$ (computable since $h_\tau \in \mathcal{L}_\star$).

Decisional case. Set $Z' := Z^{\lambda_\tau} \cdot [h_\tau(\vec{x}_0)]_\star$ and output $\mathcal{A}(\mathsf{Inst}(\vec{x}'), Z')$. If $Z = [f(\vec{x}_0)]_\star$ then

$$Z' = [\lambda_\tau f(\vec{x}_0) + h_\tau(\vec{x}_0)]_\star = [(f \circ \tau)(\vec{x}_0)]_\star = [f(\vec{x}')]_\star,$$

the real challenge for $\vec{x}'$. If Z is uniform on ch, then Z^{λ_τ} is uniform because $z \mapsto z^{\lambda_\tau}$ is a bijection of the order-p cyclic group ch (as $\gcd(\lambda_\tau, p) = 1$), so Z' is uniform and independent of $\mathsf{Inst}(\vec{x}')$. Writing p_1, p_0 for $\mathcal{A}$'s acceptance probabilities on real and random challenges,

$$\delta(\mathcal{B}; \vec{x}_0) = \mathbb{E}_\tau[p_1(\tau(\vec{x}_0))] - \mathbb{E}_\tau[p_0(\tau(\vec{x}_0))] \overset{\text{(C3)}}{=} \mathbb{E}_{\vec{x}'}[p_1] - \mathbb{E}_{\vec{x}'}[p_0] = \delta_{\mathrm{avg}}(\mathcal{A}).$$

Search case. Run $W \leftarrow \mathcal{A}(\mathsf{Inst}(\vec{x}'))$. On success $W = [f(\vec{x}')]_\star = [\lambda_\tau f(\vec{x}_0) + h_\tau(\vec{x}_0)]_\star$, so output $Y := (W \cdot [h_\tau(\vec{x}_0)]_\star^{-1})^{\lambda_\tau^{-1} \bmod p} = [f(\vec{x}_0)]_\star$. Since $\vec{x}'$ is uniform, the success probability equals $\mathrm{Succ}_{\mathrm{avg}}(\mathcal{A})$. □

Remark 5.2 (relaxing (C3))**.** When $\mathcal{D}$ makes $\tau(\vec{x})$ uniform only on a subset $\Omega \subseteq \mathbb{Z}_p^n$ (for instance $\mathbb{Z}_p^\times \times \mathbb{Z}_p$ under a multiplicative reparametrization of one coordinate), the transfer holds with the average taken over Ω. If the complement $\mathbb{Z}_p^n \setminus \Omega$ is a proper affine subvariety on which $f(\vec{x})$ takes a value the reduction can compute directly from $\mathsf{Inst}(\vec{x}_0)$ (a common case is $f(\vec{x}) = 0$ there), the reduction detects that case by an equality test on the given encodings and answers it without an oracle call, recovering full worst-case coverage.

6 Random self-reducibility: non-bilinear groups

Here $\mathcal{L}_\star = \mathcal{L}_\mathbb{G} = \langle R \rangle$ and no pairing exists, so (C2) demands the covariance defect to be a $\mathbb{Z}_p$-combination of the source exponents themselves. The motivating scalar case is discrete log: to solve g^x query the solver on g^{x+r} and subtract r [1]; its group-valued cousins fit Definition 4.1.

Proposition 6.1 (CDH and DDH)**.** *Let $R = (1, X_1, X_2)$, so $\mathcal{L}_\mathbb{G} = \langle 1, X_1, X_2 \rangle$, and $f = X_1X_2$ with challenge in $\mathbb{G}$. The translation group $\mathcal{T} = \{\vec{x} \mapsto \vec{x} + \vec{b}\}$ with $\vec{b} \leftarrow_\$ \mathbb{Z}_p^2$ satisfies* (C1)–(C3)*, so computational CDH and decisional DDH are random self-reducible.*

Proof. $X_i \circ \tau = X_i + b_i \in \mathcal{L}_\mathbb{G}$ gives (C1). For (C2), $f \circ \tau = (X_1 + b_1)(X_2 + b_2) = f + (b_2X_1 + b_1X_2 + b_1b_2)$, so $\lambda_\tau = 1$ and $h_\tau = b_2X_1 + b_1X_2 + b_1b_2 \in \langle 1, X_1, X_2 \rangle = \mathcal{L}_\star$. Translation gives exact uniformity, (C3). Explicitly the reduction returns $[f(\vec{x}_0)] = W \cdot [x_1]^{-b_2}[x_2]^{-b_1}[1]^{-b_1b_2}$ in the search case. □

With $R = (1, X_1, \ldots, X_n)$ we have $\mathcal{L}_\star = V_{\leqslant 1}$ (affine polynomials), and translating a degree-d monomial leaves monomials of degree at most $d-1$; hence (C2) holds under translation exactly when $\deg f \leqslant 2$. Without a pairing, degree 2 is the ceiling: a product target is the most that is randomizable, and it is.

7 Random self-reducibility: Type 1

Here $\mathcal{L}_\star = \mathcal{L}_{\mathbb{G}_T} = \langle R \cdot R \cup T \rangle$: the pairing supplies all products of source exponents, so the covariance defect may use any degree-2 combination of them.

Proposition 7.1 (BDDH)**.** *Let $R = (1, X_1, X_2, X_3)$, $T = (1)$, so $\mathcal{L}_{\mathbb{G}_T} = V_{\leqslant 2}$, and $f = X_1X_2X_3$ with challenge in $\mathbb{G}_T$. Translation satisfies* (C1)–(C3)*, so decisional and computational BDDH are random self-reducible.*

Proof. $X_i \circ \tau = X_i + b_i \in \mathcal{L}_\mathbb{G} = \langle R \rangle$ gives (C1). Expanding,

$$f \circ \tau = X_1X_2X_3 + \left[b_3X_1X_2 + b_2X_1X_3 + b_1X_2X_3 + (\deg \leqslant 1)\right],$$

so $\lambda_\tau = 1$ and h_τ has total degree at most 2, hence lies in $V_{\leqslant 2} = \mathcal{L}_\star$; each of its monomials is realized by a single pairing, for instance $[x_1x_2]_T = e([x_1]_1, [x_2]_2)$. Translation gives (C3). $\square$

With coordinate source $R = (1, X_1, \dots, X_n)$ we get $\mathcal{L}_\star = V_{\leqslant 2}$, so the same degree count gives translation-randomizability for every f with $\deg f \leqslant 3$. The pairing has raised the ceiling from 2 (no pairing) to 3.

8 Random self-reducibility: Type 2

Here $\mathcal{L}_1 = \langle R \cup S\rangle$ and $\mathcal{L}_T = \langle (R \cup S) \cdot S \cup T\rangle$: relative to Type 3 the homomorphism ψ adds the right-hand products $S \cdot S$ to the target space. This extra room is what makes the following target randomizable.

Proposition 8.1. *In a Type-2 group let $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, $T = (1)$, and $f = X_1X_2^2$ with challenge in $\mathbb{G}_T$. Then $f \notin \mathcal{L}_\star$ (so the assumption is generically hard), and translation satisfies* (C1)–(C3)*; the problem is random self-reducible.*

Proof. Here $\mathcal{L}_T = \langle (R \cup S) \cdot S\rangle = \langle 1, X_1, X_2, X_1X_2, X_2^2\rangle$, and $X_1X_2^2$ is not in this span, giving generic hardness. Under $\tau\colon \vec{x} \mapsto \vec{x} + \vec{b}$ one has $X_1 \circ \tau = X_1 + b_1 \in \mathcal{L}_1 = \langle 1, X_1\rangle$ and $X_2 \circ \tau = X_2 + b_2 \in \mathcal{L}_2 = \langle 1, X_2\rangle$, which is (C1). Expanding,

$$f \circ \tau = (X_1 + b_1)(X_2 + b_2)^2 = f + \underbrace{2b_2X_1X_2 + b_2^2X_1 + b_1X_2^2 + 2b_1b_2X_2 + b_1b_2^2}_{h_\tau},$$

so $\lambda_\tau = 1$ and every monomial of h_τ lies in $\mathcal{L}_\star$. The term X_2^2 is realized using ψ: $[x_2^2]_T = e\big(\psi([x_2]_2), [x_2]_2\big)$. Translation gives (C3). $\square$

The point is precisely the X_2^2 term of h_τ: it is a product of two $\mathbb{G}_2$ exponents, which the pairing can form only after ψ has moved one of them into $\mathbb{G}_1$. In Type 1 this term is free; in Type 3 it is unavailable, as the next section shows.

9 Random self-reducibility: Type 3

Here $\mathcal{L}_1 = \langle R\rangle$, $\mathcal{L}_2 = \langle S\rangle$, and $\mathcal{L}_T = \langle R \cdot S \cup T\rangle$: the target space contains only cross products. This is the most restrictive setting, and reparametrizations must respect which coordinates are reachable in the challenge group.

Proposition 9.1 (co-CDH)**.** *In a Type-3 group let $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, and $f = X_1X_2$ with challenge in $\mathbb{G}_1$. Then $f \notin \mathcal{L}_\star$ (generically hard), and the reparametrization $\tau\colon (x_1, x_2) \mapsto (\alpha x_1,\ x_2 + b_2)$ with $\alpha \leftarrow_\$ \mathbb{Z}_p^\times$, $b_2 \leftarrow_\$ \mathbb{Z}_p$ satisfies* (C1),(C2) *and makes $\tau(\vec{x})$ uniform on $\mathbb{Z}_p^\times \times \mathbb{Z}_p$; the problem is random self-reducible, with the slice $x_1 = 0$ handled directly.*

Proof. Here $\mathcal{L}_\star = \mathcal{L}_1 = \langle 1, X_1\rangle$, which contains no $\mathbb{G}_2$ variable; and $X_1X_2 \notin \langle 1, X_1\rangle$, giving hardness. For (C1), $X_1 \circ \tau = \alpha X_1 \in \mathcal{L}_1$ and $X_2 \circ \tau = X_2 + b_2 \in \mathcal{L}_2 = \langle 1, X_2\rangle$. For (C2), $f \circ \tau = \alpha X_1(X_2 + b_2) = \alpha f + \alpha b_2 X_1$, so $\lambda_\tau = \alpha \in \mathbb{Z}_p^\times$ and $h_\tau = \alpha b_2 X_1 \in \mathcal{L}_\star$. The image

$\tau(\vec{x}) = (\alpha x_1, x_2 + b_2)$ is uniform on $\mathbb{Z}_p^\times \times \mathbb{Z}_p$ for every fixed $x_1 \neq 0$. If $x_1 = 0$ then $f(\vec{x}) = 0$ and the answer is the identity of $\mathbb{G}_1$; the reduction detects this by testing $[x_1]_1 = [0]_1$ and outputs the identity, per Remark 5.2. A multiplicative reparametrization of x_1 is forced because an additive shift $x_1 \mapsto x_1 + b_1$ would put $b_1 X_2$ into h_τ, and $X_2 \notin \mathcal{L}_1$ in Type 3. $\square$

The next result shows the Type-3 restriction genuinely bites: the target of Proposition 8.1, self-reducible in Type 2, is not affine-randomizable in Type 3.

Proposition 9.2 (a Type-2 versus Type-3 separation)**.** *In a Type-3 group let $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, $T = (1)$, and $f = X_1 X_2^2$ with challenge in $\mathbb{G}_T$. The assumption is generically hard, and no $\tau \in \mathrm{AGL}_2(\mathbb{Z}_p)$ satisfies* (C1)–(C3)*; hence the affine reduction of Theorem* 5.1 *does not apply, in contrast to Type 2 (Proposition* 8.1*).*

Proof. Hardness holds as $\mathcal{L}_\star = \mathcal{L}_T = \langle R \cdot S \rangle = \langle 1, X_1, X_2, X_1X_2 \rangle$ and $X_1X_2^2 \notin \langle 1, X_1, X_2, X_1X_2 \rangle$. Suppose $\tau \in \mathrm{AGL}_2(\mathbb{Z}_p)$ satisfies (C1). Condition (C1) for $r = X_1$ forces the first coordinate map $\tau_1 = X_1 \circ \tau$ to lie in $\mathcal{L}_1 = \langle 1, X_1 \rangle$, that is $\tau_1 = e_1 X_1 + e_0$ with no X_2; likewise (C1) for $s = X_2$ forces $\tau_2 = a X_2 + a_0$ with no X_1. Invertibility requires $e_1, a \in \mathbb{Z}_p^\times$. Then

$$f \circ \tau = (e_1 X_1 + e_0)(a X_2 + a_0)^2 = (e_1 X_1 + e_0)\left(a^2 X_2^2 + 2 a a_0 X_2 + a_0^2\right).$$

The coefficient of the monomial X_2^2 (which carries no X_1) is $e_0 a^2$. Since $X_2^2 \notin \mathcal{L}_\star$ and $\lambda_\tau f = \lambda_\tau X_1 X_2^2$ contributes no pure X_2^2 term, condition (C2) forces $e_0 a^2 = 0$, hence $e_0 = 0$. But then $\tau_1 = e_1 X_1$, so $\tau(\vec{x})_1 = e_1 x_1$, which cannot be uniform on $\mathbb{Z}_p$ for a fixed x_1 (it never realizes uniformity across the value 0), so (C3) fails. Therefore no τ meets (C1)–(C3) simultaneously. $\square$

The obstruction is exactly the one flagged after Proposition 8.1: the covariance defect necessarily contains X_2^2, a product of two $\mathbb{G}_2$ exponents. In Type 2 the homomorphism ψ places X_2^2 in $\mathcal{L}_T$ and translation works; in Type 3 it is generically uncomputable, and forcing it out of the defect collapses the reparametrization on the $\mathbb{G}_1$ coordinate. Proposition 9.2 rules out the affine technique, which is the general tool behind all positive results here; a proof that no reduction whatsoever exists would additionally require a lower-bound argument such as a meta-reduction, but the structural gap is genuine.

10 Discussion

Across the four settings the reduction of Theorem 5.1 is one and the same; only the admissibility conditions move, and they move only because the computable target space $\mathcal{L}_\star$ grows with the available products:

$$\text{none} \subsetneq R \cdot S \text{ (Type 3)} \subsetneq R \cdot S \cup S \cdot S \text{ (Type 2)} \subseteq R \cdot R \text{ (Type 1, single group)}.$$

A larger $\mathcal{L}_\star$ makes the covariance condition (C2) easier to meet, so the affine-self-reducible subclass is largest in Type 1 and smallest in Type 3, with the non-bilinear case the degenerate product-free floor. Proposition 9.2 witnesses the strict gap between Type 2 and Type 3.

Self-reducibility is orthogonal to hardness. The master theorem of [3, 4] makes an assumption hard exactly when $f \notin \mathcal{L}_\star$, whereas Theorem 5.1 needs the defect $f \circ \tau - \lambda_\tau f$ to lie *in* $\mathcal{L}_\star$. These constrain different objects, so hard yet self-reducible assumptions abound:

CDH and DDH (non-bilinear), BDDH (Type 1), $X_1X_2^2$ (Type 2), and co-CDH (Type 3) are all of this kind. Quantitatively, Bauer, Fuchsbauer, and Loss [6] show that in the algebraic group model every member of the Uber family is implied by the q-discrete-logarithm assumption, with q governed by the degrees of the defining polynomials, and Rotem [7] refines q to the maximal degree in which a single variable occurs and gives improved bilinear reductions.

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